

Interpolation

- Interpolation is a method of estimating unknown values that fall within the range of a given set of known data points.
- It is used when:
 - Function is difficult or impossible to express analytically
 - Only discrete data is available (e.g., from experiments, tables)
- The goal is to construct an approximate function $f(x)$ such that $f(x_i)=y_i$ for known points (x_i, y_i) , and estimate $f(x)$ for intermediate values of x .

Feature	Interpolation	Extrapolation
Definition	Estimating a value within known data range	Estimating a value outside the known range
Accuracy	Generally more accurate	Less accurate and more uncertain
Data requirement	Uses nearby known points	Projects trend beyond known data
Example	Estimate value at $x=2.5$ using data from $x=1,2,3$	Estimate value at $x=6$ using data from $x=1,2,...,5$

Newton's Forward Interpolation (For equal intervals, near the beginning of the table)

- Used when x lies **near the beginning** of the data table.
- Assumes **equal intervals**: $x_{i+1} - x_i = h$ (constant).
- Based on **forward difference table** using symbol Δ .
- Approximates unknown values by using a **polynomial expansion** of the known values starting from y_0 .
- Involves a variable $u = \frac{x-x_0}{h}$ to generalize the formula.
- Effective when data points are **closely and equally spaced**.

Newton's Backward Interpolation (For equal intervals, near the end of the table)

- Used when x lies **near the end** of the data table.
- Also assumes **equal spacing** between x -values.
- Based on the **backward difference table** using symbol ∇ .
- Uses a polynomial formed in terms of y_n (last known value).
- Introduces $u = \frac{x-x_n}{h}$, where x_n is the last known value.
- Useful when approximating values **after the last known data point** or very close to it.

1. Newton's Divided Differences

Newton's Divided Difference method is a recursive formula to construct an interpolating polynomial that passes through a given set of data points.

◆ Form of the Polynomial:

$$P(x) = f[x_0] + f[x_0, x_1](x - x_0) + f[x_0, x_1, x_2](x - x_0)(x - x_1) + \dots$$

- $f[x_0]$, $f[x_0, x_1]$, etc. are divided differences computed recursively.

◆ Divided Difference Formula:

$$f[x_i, x_{i+1}] = \frac{f(x_{i+1}) - f(x_i)}{x_{i+1} - x_i}$$

$$f[x_i, x_{i+1}, x_{i+2}] = \frac{f[x_{i+1}, x_{i+2}] - f[x_i, x_{i+1}]}{x_{i+2} - x_i}$$

◆ Advantages:

- Works well when data points are added incrementally.
- Efficient for unequally spaced data.

2. Lagrange's Interpolation

Lagrange interpolation constructs a single polynomial that passes through all given points using Lagrange basis polynomials.

◆ Form of the Polynomial:

$$P(x) = \sum_{i=0}^n y_i \cdot L_i(x)$$

$$L_i(x) = \prod_{\substack{j=0 \\ j \neq i}}^n \frac{x - x_j}{x_i - x_j}$$

- $L_i(x)$ is the Lagrange basis polynomial.
- y_i are the given function values at x_i .

◆ Advantages:

- Simple to implement.
- Useful when the entire data set is known in advance.